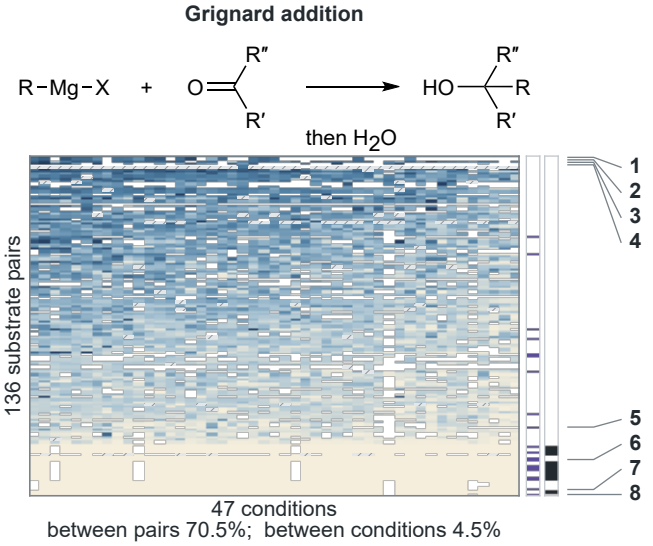
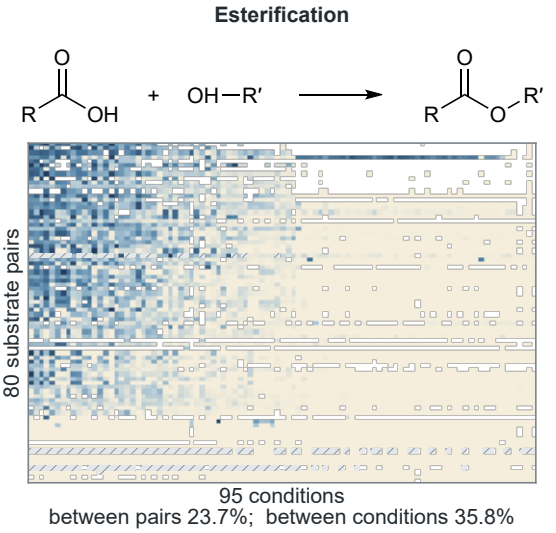
**a Grignard yields follow the substrate pair; esterification yields follow the conditions**



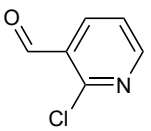

amino or N–H on carbonyl | reagent never gave product | no numerical yield | no record

yield, % 0 50 100

**b Selected carbonyl partners: amino or N–H groups give little product**

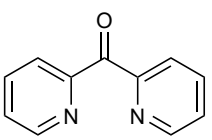
highest pair means

**1**



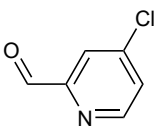
+ 4-$FC_6H_4MgBr$
**pair mean 75%**

**2**



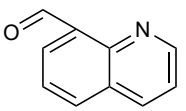
+ 2-$MeC_6H_4MgBr$
**pair mean 73%**

**3**



+ 4-$MeC_6H_4MgBr$
**pair mean 72%**

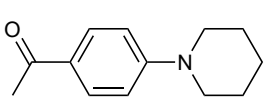
**4**



+ 4-$MeC_6H_4MgCl$
**pair mean 72%**

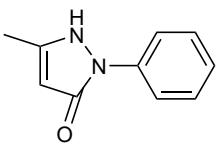
amino or N–H partners

**5**



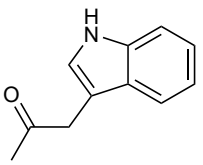
+ 2-$MeOC_6H_4MgBr$
**pair mean 14%**

**6**



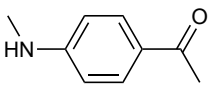
+ 3-$MeOC_6H_4MgBr$
**pair mean 0%**

**7**



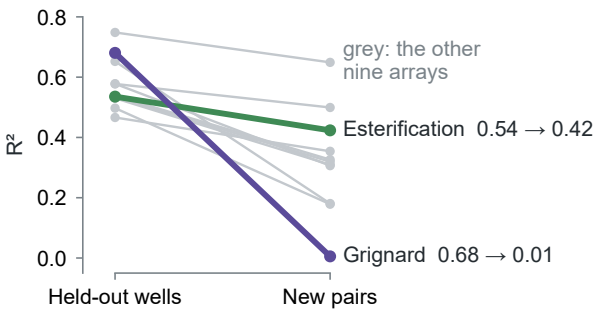
+ 4-$MeC_6H_4MgBr$
**pair mean 0%**

**8**



+ 4-$Me_2NC_6H_4MgBr$
**pair mean 0%**

**c Grignard keeps almost none of its accuracy**




**d Seen substrates do not rescue Grignard**

